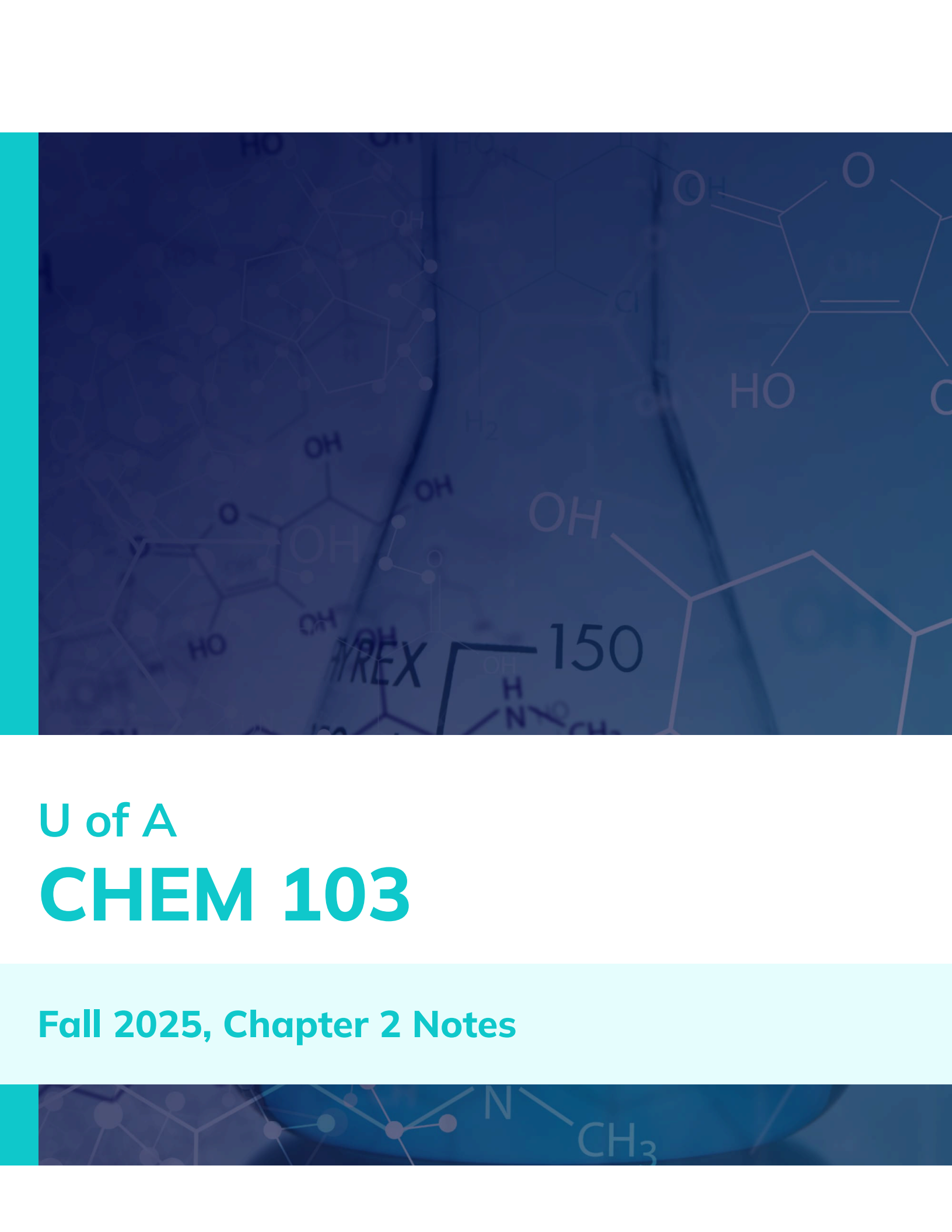




U of A **CHEM 103**

Fall 2025, Chapter 2 Notes

Table of Contents

Chapter 2. Unit 1. Modern Atomic Theory

2.1. Electromagnetic Radiation and the Nature of Light

2.1.1. Electromagnetic Spectrum

2.1.2. Photon Equations

2.1.3. Photon Equations (Cntd.)

2.1.4. Example

2.1.5. Example

2.1.6. Practice

2.1.7. Practice

2.2. Blackbody Radiation

2.2.1. Blackbody Radiation

2.3. The Photoelectric Effect

2.3.1. The Photoelectric Effect

2.3.2. The Photoelectric Effect Equation

2.3.3. Example

2.3.4. Practice

2.3.5. Practice

2.4. Understanding the Bohr Model

2.4.1. The Basics of Absorption and Emission Spectra

2.4.2. Bohr's Atomic Model

2.4.3. Absorption vs Emission of a Photon

2.4.4. Example

2.4.5. Example

2.5. Bohr Model Equations

2.5.1. Bohr Model Equations

2.5.2. Example

2.5.3. Example

2.5.4. Practice

2.6. How Atomic Spectra Relates to Electronic Transitions + Other Applications of the Bohr Model

2.6.1. The Bohr Model and Atomic Line Spectra

2.6.2. Relating Electronic Transitions to Emission Spectra

2.6.3. Example

2.6.4. Practice

2.6.5. Practice

2.7. Wave-Particle Duality & the de Broglie Equation

2.7.1. Wave-Particle Duality

2.7.2. Example

2.7.3. Practice

2.7.4. Practice

2.8. Schrodinger's Equation

2.8.1. Schrodinger's Equation

2.9. Heisenberg Uncertainty Principal

2.9.1. The Heisenberg Uncertainty Principle

2.9.2. Example

2.9.3. Practice

2. Unit 1. Modern Atomic Theory

2.1 Electromagnetic Radiation and the Nature of Light

2.1.1

Electromagnetic Spectrum

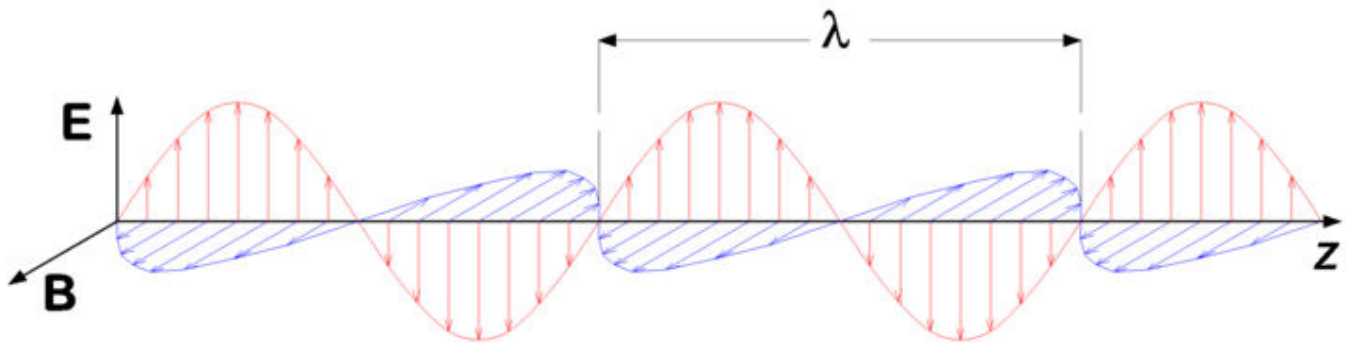
Up until this point in history, scientists were unsure if light shared the properties of waves or particles...there was some evidence for both!

The discovery of the electromagnetic spectrum showed the different wavelengths and frequencies that electromagnetic radiation could exist at. Since the different **types of radiation in this spectrum behave as waves**, and visible light is a part of this spectrum, it became clear to scientists that **light acts as waves!**

Photon Equations

Electromagnetic Radiation Waves

- Light (or electromagnetic radiation) behaves as a wave
- A wave is a self-propagating transverse oscillation of electric and magnetic fields



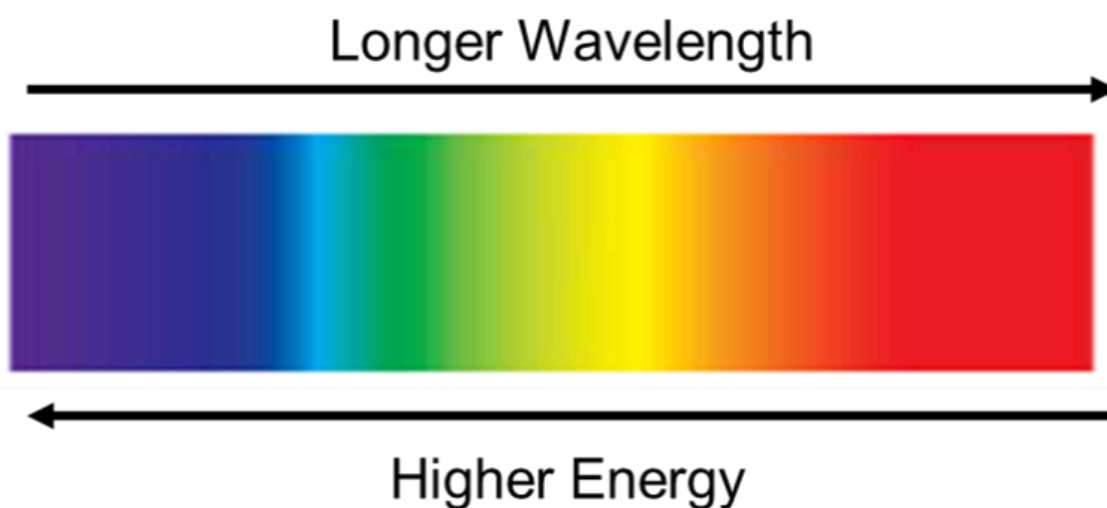
where E is the electric field and B is the magnetic field

- To describe waves, we can use the following terms:
 - **Amplitude (A)**: the height of a wave crest or depth of a trough
 - **Wavelength (λ)**: The distance a wave travels in one cycle (**m**)
 - **Frequency (ν)**: cycles per second (s^{-1} or units **Hz**)
 - **Energy (E)**: units are Joules (**J**)

Energy of a Photon Equations

$$E = \frac{hc}{\lambda} = h\nu$$

where *h* is Planck's constant: $6.626 \times 10^{-34} \text{ Js}$
and *c* is the speed of light: $3.0 \times 10^8 \text{ m/s}$



i WIZE TIP

You should **memorize the value for the speed of light (c) = $3.0 \times 10^8 \text{ m/s}$** as that value is not always provided on exams.

💡 WIZE CONCEPT

Where will I see this on an exam?

Questions on this topic tend to arise in the form of a calculation. You will be asked to determine the energy or wavelength of a given ray of light.

Photon Equations (Cntd.)

Key Relationships Between Variables in Photon Equations

$$E = \frac{hc}{\lambda}$$

- Are the E and λ of a photon directly or inversely proportional? _____ proportional

$$E = h\nu$$

- Are the E and ν of a photon directly or inversely proportional? _____ proportional

$$\frac{hc}{\lambda} = h\nu$$

- Are the wavelength and frequency of a photon directly or inversely proportional?
_____ proportional

Summary

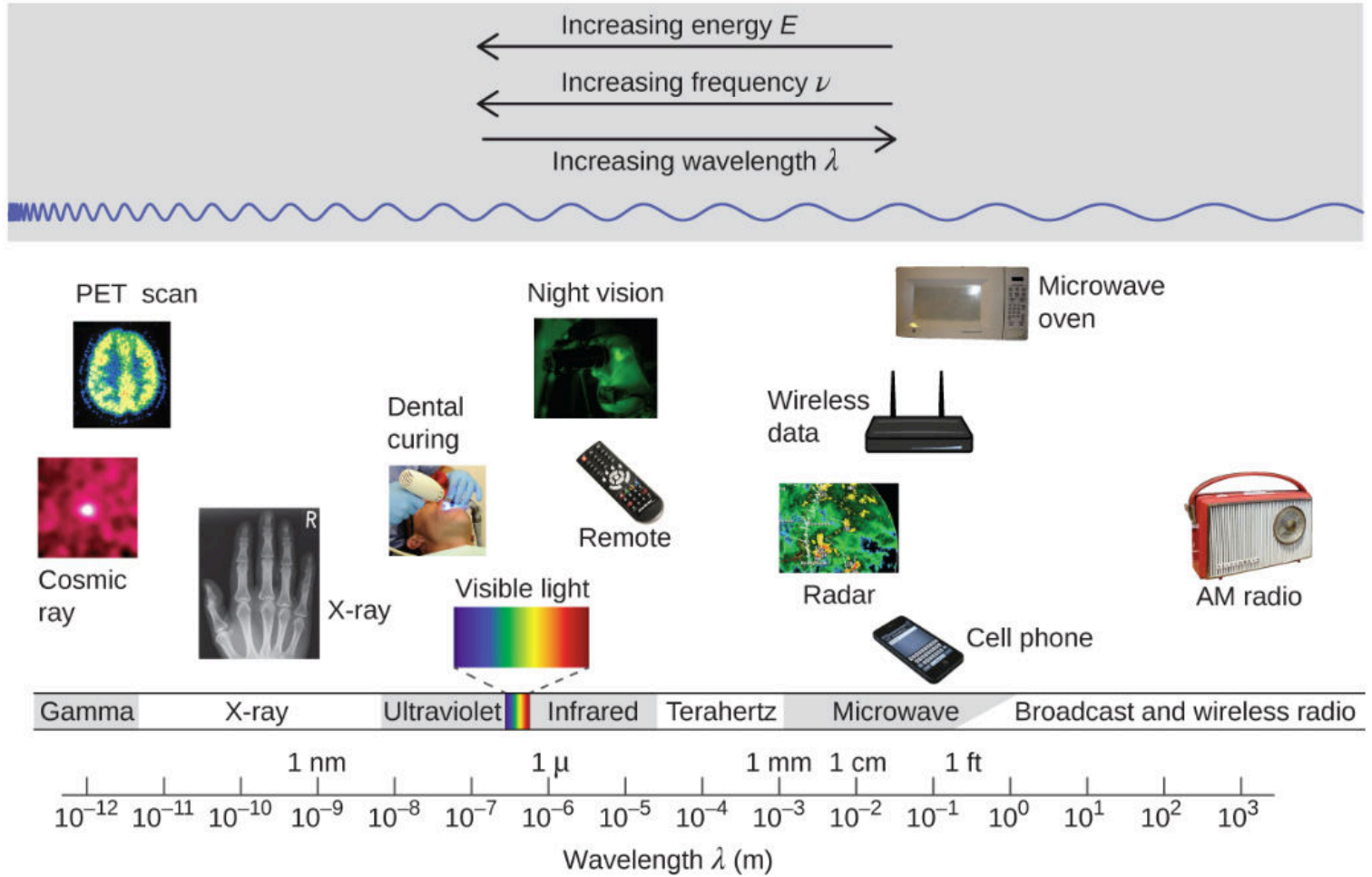


Photo by Rice University/ CC BY

2.1.4 Example

Example: Calculating Energy of a Photon Given Wavelength

A photon of light was found to emit light at 723 nm. How much energy is associated with this light?

$$h = 6.626 \times 10^{-34} \text{ Js}$$

! WATCH OUT!

Don't forget that for the photon equations **E must in joules (J)** and **λ (wavelength) must be in m (meters) not nm (nanometers)!**

- If you are given a wavelength value in nm, you would have to convert it to m before plugging it into an equation.

Remember: $1\text{nm}=1\times 10^{-9}\text{m}$

Example: Calculating Frequency of a Photon Given Energy

Calculate the frequency of a photon with an energy of 236 kJ.

$$h = 6.626 \times 10^{-34} \text{ Js}$$

Practice: Different Types of Electromagnetic Radiation

Rank the following types of electromagnetic radiation from lowest to highest energy.

Radio Waves, Infrared Radiation, X-rays, Microwaves, Visible Light, UV Radiation

X-rays < Infrared Radiation < Microwaves < UV Radiation < Visible Light < Radio Waves

☐

Radio Waves < Microwaves < Infrared Radiation < Visible Light < UV Radiation < X-rays

☐

Radio Waves < Infrared Radiation < Microwaves < Visible Light < UV Radiation < X-rays

☐

Radio Waves < Infrared Radiation < Visible Light < Microwaves < UV Radiation < X-rays

☐

Practice: Calculating Frequency of a Photon Given Wavelength

A photon of light has a wavelength of 700 nm. What is its frequency?

$4.29 \times 10^{-11} \text{ s}^{-1}$

☐

$6.72 \times 10^{12} \text{ s}^{-1}$

☐

$4.29 \times 10^{14} \text{ s}^{-1}$

☐

$7.23 \times 10^{16} \text{ s}^{-1}$

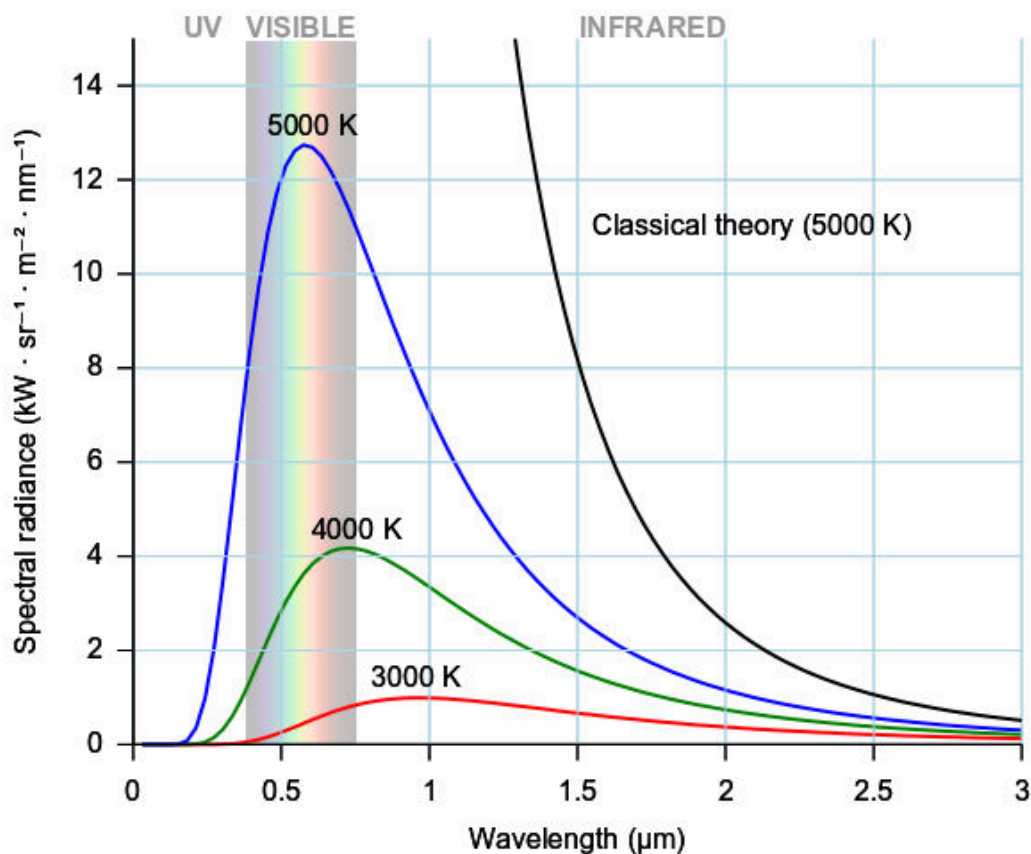
☐

2.2 Blackbody Radiation

2.2.1

Blackbody radiation

Classical Problem:



Issues:

1. Classical Theory does not do a good job predict the emission of bodies in the infrared region, especially for cooler bodies.
2. Near the UV-range classical theory predicts an infinite increase in radiance which means an infinite amount of light being emitted. This is a clear contradiction of energy conservation, a BIG ISSUE! Known as the UV-Catastrophe

Solution:

1. Max Plank discovered that if the restricted energy that can be released or absorbed by atoms into discrete energy packets he could successfully predict the experimental observations.

Results:

1. The smallest quantity of energy that can be emitted or absorbed as electromagnetic radiation, is called a quanta

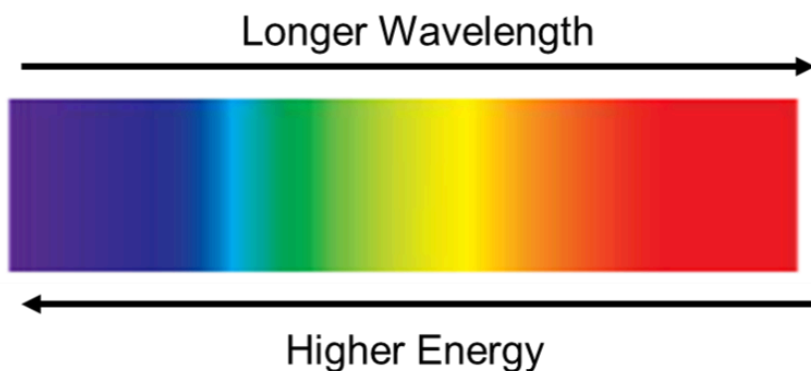
The energy of a single quantum is:

$$E = hv = \frac{hc}{\lambda}$$

E = energy of the photon in J

$h = 6.626 \times 10^{-34} \text{ Js}$ (Planck's Constant)

ν = frequency in s^{-1}



i WIZE TIP

You should **memorize the value for the speed of light (c) = $3.0 \times 10^8 \text{ m/s}$** as that value is not always provided on exams.

 WIZE CONCEPT

Where will I see this on an exam?

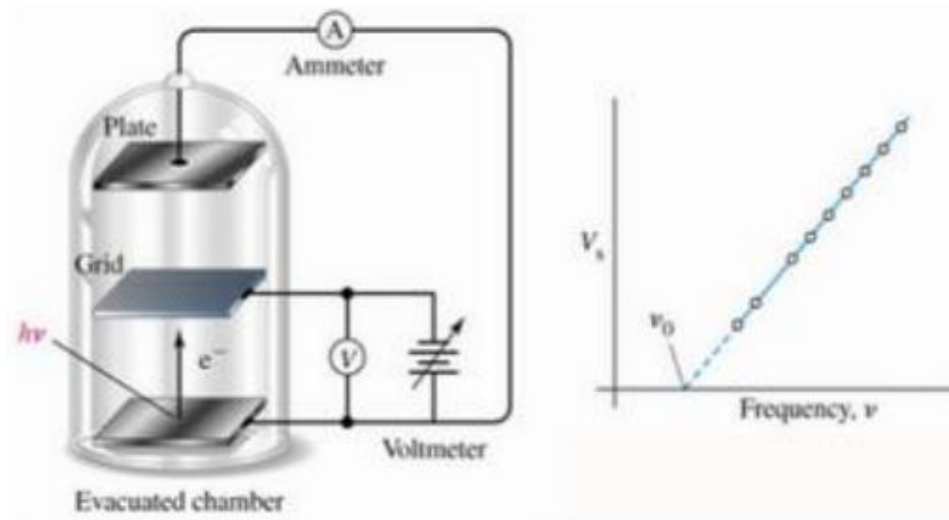
Questions on this topic tend to arise in the form of a calculation. You will be asked to determine the energy or wavelength of a given ray of light.

2.3 The Photoelectric Effect

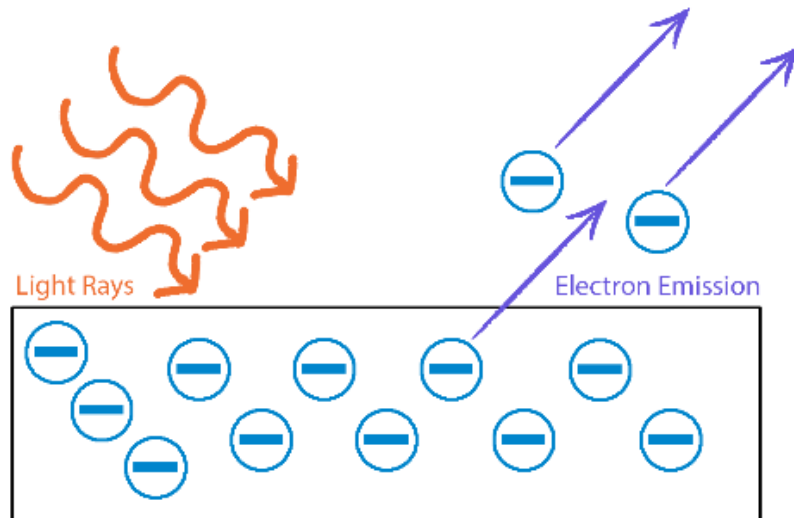
2.3.1

The Photoelectric Effect

In 1905, Einstein conducted his **photoelectric effect** experiment.

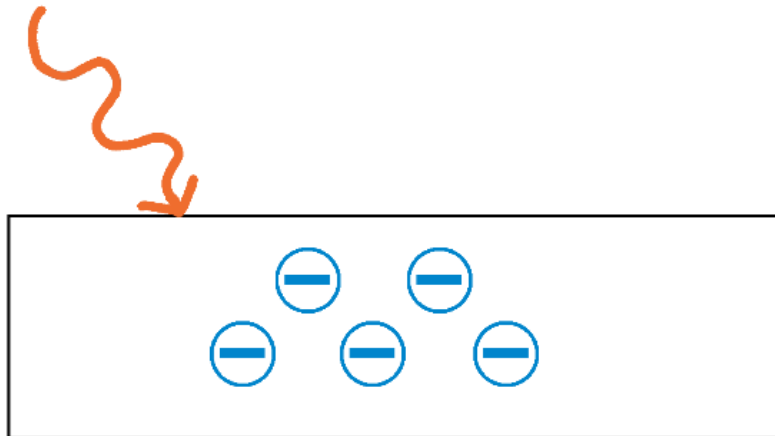


- The apparatus is made of a metal grid in which light of different intensities and frequencies was shown onto it.
- The ammeter (A) measures current (the **number of electrons emitted**).
- The voltmeter (V) measures the stopping potential (the **energy of the electrons emitted**).

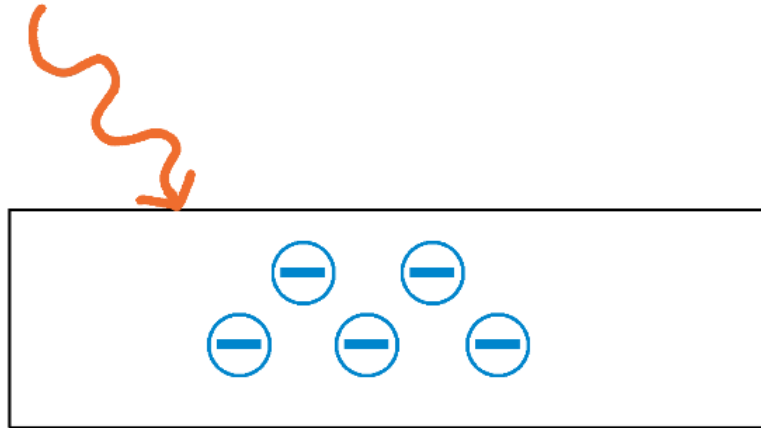


Experimental Findings

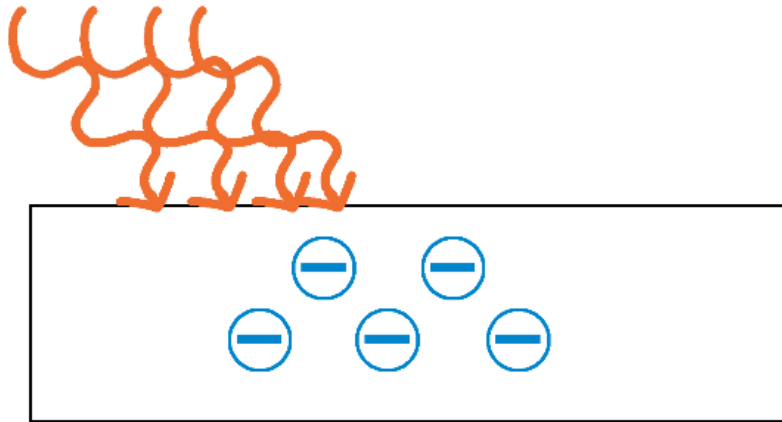
1) 1 photon with **sufficient energy** to eject an electron:



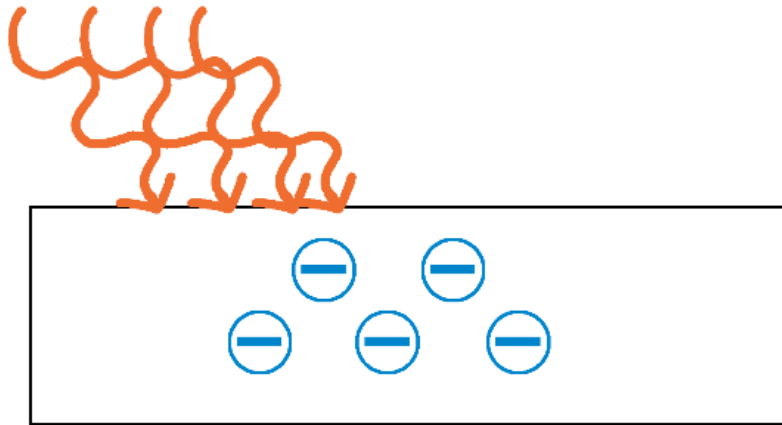
2) 1 photon with **insufficient energy** to eject an electron



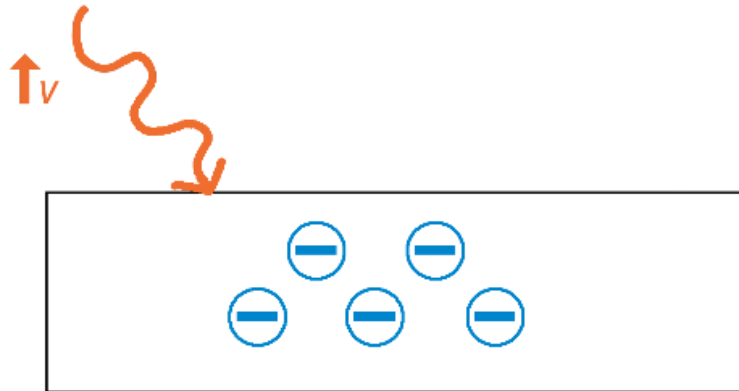
3) Multiple photons hitting the metal, **individually they have insufficient energy** to eject an electron but if you add up the their energy they have more energy than required to eject an electron



4) Multiple photons, **each with sufficient energy** needed to eject an electron hitting the metal



5) A photon with an **increased frequency** than the photons with minimum sufficient energy to eject an electron



Summary of the Findings:

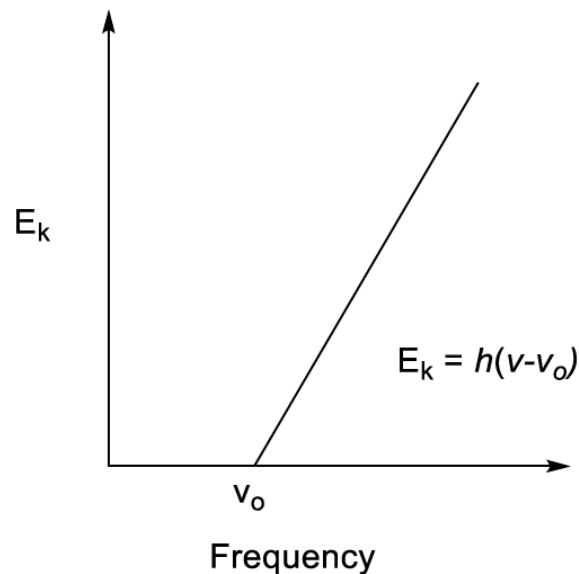
- **Electrons must be ejected by 1:1, photon : electron collisions**
- **Increased intensity** of the light (**more photons** for a given part of the light beam), **increased the number of electrons emitted** (if **each of the photons had sufficient energy** to eject electrons)
- **Increased frequency** of the light, **increased the energy of the electrons emitted**

Before this experiment, people thought of light only as WAVES, but **Einstein showed that light could also be thought of as particles (photons).**

- When particle A hits particle B, particle A transfers its momentum to particle B. This is not the case for waves!
- The photoelectric effect showed that light can behave like particles since photons of light can hit a solid substance and cause electrons to be emitted

Photoelectric Effect Equation

This experiment shows that light can behave like a particle “knocking” electrons out of a material.



- The **y-axis** shows the **kinetic energy (E_k) of the emitted electron**
- The **x-axis** shows the **frequency of the photon** that is hitting the metal's surface
 - When the **frequency is too low**, the energy of the photon is too low and an **electron is not emitted**
 - When the **frequency reaches the threshold frequency**, there is **enough energy to emit an electron**
 - At even **higher frequencies**, the **kinetic energy (E_k) of the electron is increased further!**

The energy corresponding to v_0 is called the '**binding energy**' and represents the amount of **energy required to remove the electron, E_{Binding}**

$$E_k \text{ electron being ejected} = E_{\text{photon}} - E_{\text{binding}}$$

$$\text{where } E_{\text{photon}} = hv = \frac{hc}{\lambda}$$

and $E_{k_{electron}} = \frac{1}{2}mv^2$
and $E_{binding}$ = minimum amount of energy needed to eject an electron

i WIZE TIP

The **binding energy for each metal is unique.**

Example: The binding energy of iron and gold are different.

The **binding energy for a specific metal is constant** so if you are given the binding energy for iron in a question, you can use that for part a) b) c) for example.

Does an electron get displaced when $E_{photon} >$ or $<$ than $E_{binding}$?

Example: Ejecting an Electron

When lithium metal is irradiated by a laser with a wavelength of 400 nm no electrons are ejected from the surface. If you want to ionize lithium which of the following types of radiation should you use?

- a) UV-Laser
- b) IR-Radiation
- c) Red Light Laser
- d) Green Light Laser
- e) Radio Wave

WIZE TIP

When you see a question that mentions **electrons being ejected** from the surface of a metal, you know the question is going to have something to do with the **photoelectric effect**!

Practice: No Electrons are Ejected

A metal surface is irradiated by light from a laser with a wavelength of 492 nm. No electrons are measured leaving the surface. What is the most likely explanation?

All of the electrons which are easy to remove have already been removed

☐

The frequency of the photons is less than the threshold frequency of the metal

☐

The energy of the photons is greater than the threshold value

☐

The light is not irradiating the surface at the correct angle

☐

None of the above

☐

Practice: Calculate the Binding Energy

An element is bombarded with a beam of X-rays of wavelength 0.75664nm. Upon irradiation, electrons are ejected with a velocity of 2.35×10^7 m/s. The mass of an electron is 9.11×10^{-31} kg.

What is the binding energy for the element?

$$h = 6.626 \times 10^{-34} \text{ Js}$$

☐ $1.12 \times 10^{-14} J$

☐ $2.46 \times 10^{-17} J$

☐ $1.12 \times 10^{-17} J$

☐ $0 J$

2.4 Understanding the Bohr Model

2.4.1

The Basics of Absorption and Emission Spectra

Each atom has a unique absorption and emission spectrum.

Continuous Spectrum



Emission Lines



Absorption Lines



Absorption Spectra

- We get an absorption spectrum by transmitting electromagnetic radiation (light) through a substance.
- The **dark bands** in the spectra represent all of the **specific wavelengths of photons absorbed** by the atom's electrons

Absorption Lines



Emission Spectra

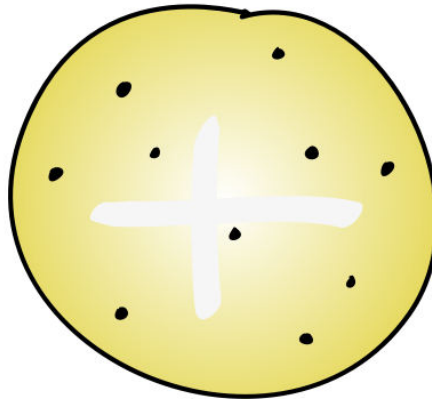
- We get an emission spectrum by measuring the electromagnetic radiation (light) that is emitted from a substance
- The background is black and the **coloured bands** represent the **wavelengths of photons that were emitted** from this atom's electrons

Emission Lines



Bohr's Atomic Model

Before the Bohr model, scientists thought that electrons just assumed arbitrary energies and could be found anywhere in the atom.



If the pre-Bohr model was right, what would the emission spectra look like? Would they have lines or would they be continuous spectra? _____

Niels Bohr realized that the existing model at the time was inconsistent with emission spectra data... if electrons could be anywhere and have any energy, why did each element have a unique emission spectra with photons of light being emitted with specific energies and wavelengths?

Bohr realized that the **electrons must have discrete energies** in order to explain the emission spectra with photons with specific amounts of energy being released.

Bohr's Atomic Model Details

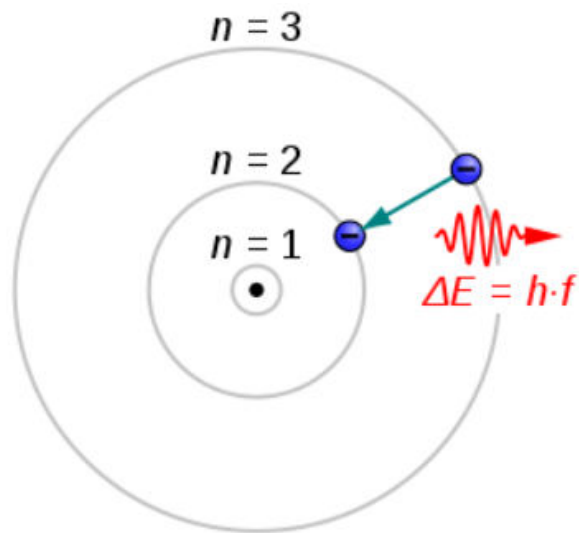


Photo by JabberWok / CC BY

- Electrons orbit nucleus at **fixed distances** and in **circular paths** (like how planets orbit the sun!)
- Energy electrons have are **quantized** (only specific energies are allowed)
- Shell's distance from nucleus determines energy ($n=1, 2, 3$, etc, **higher n =higher energy shell**)
- **Gap between levels shrinks as n grows**
- Bohr model is only accurate for atoms with **1 electron** ("**Bohr atoms**")
 - The Bohr model explains why only photons with certain energies are observed in the emission spectra!

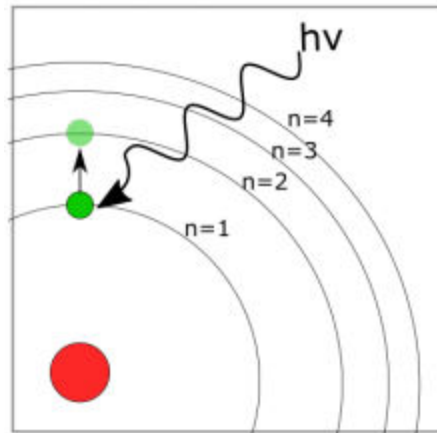
Analogy: Staircase vs Ramp



- **Ramp**-> if you're walking up a ramp, you could be 12cm, 17cm, or 20cm above the ground. (Pre-Bohr Model)
- **Staircase**-> if each step is 17cm high, you can never be standing 12 cm or 20cm above the ground (Bohr Model)
 - The **stairs are discrete units** just like **shells in an atom are discrete units**.
 - And just like how we can only be **certain heights above the ground when we take the stairs**, **electrons can only be in certain shells with certain energies (can't be between two shells)**

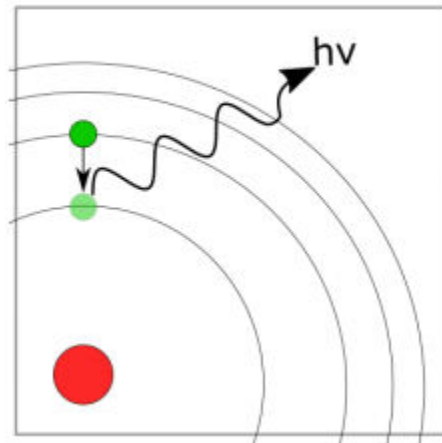
Absorption vs Emission of a Photon

Absorption



- Atom **absorbs photon** with **energy** ($E=h\nu$) **equal to gap between levels**
- Electron is promoted to **higher energy level** (ex: $n = 1 \rightarrow n = 2$)
- **Greater Δn = higher energy**

Emission



- Atom **emits photon** with **energy equal to gap between levels**
- Electron relaxes to **lower energy level** (ex: $n = 4 \rightarrow n = 2$)

Example: Understanding Absorption Spectra and the Bohr Model

To collect an absorption spectra of a gaseous mixture a gas is irradiated with light from each wavelength in a given range and the amount of light which passes through the sample is recorded. Some of the light at certain wavelengths is absorbed by the sample and these wavelengths become peaks in the absorption spectra.

1) An absorption event occurs when...

- a) The atom is destroyed
- b) The nuclear charge increases by 1
- c) An isotope ejects an electron
- d) An electron is excited from one energy level to a higher energy level
- e) None of the above

2) The wavelength of light which is absorbed corresponds with ...

- a) The difference in energy between the energy levels which the electrons occupy
- b) The energy level of the orbit the electron was initially in
- c) The energy level of the orbit the electron ends up in
- d) The distance between the two orbits the electron moves between
- e) None of the above

Example: Electron Transitions

An electron in the $n = 4$ shell of an excited hydrogen atom relaxes to its ground state. Which of the following is true?

- a) A photon is absorbed and with energy equal to the energy of the ground state
- b) A photon is absorbed with energy slightly less than the difference in energy between $n = 4$ and $n = 1$
- c) A photon is absorbed with energy equal to the difference in energy between $n = 4$ and $n = 1$
- d) A photon is released with energy equal to the energy of the ground state
- e) A photon is released with energy slightly less than the difference in energy between $n = 4$ and $n = 1$
- f) A photon is released with energy equal to the difference in energy between $n = 4$ and $n = 1$

2.5 Bohr Model Equations

2.5.1 Bohr Model Equations

Bohr Model Equations

There are a few important equations we should be familiar with that are related to the Bohr model. **These equations can only be used for one electron systems!** Recall that **Bohr's model** was only able to explain species with **1 electron**.

Calculating the Energy of a Specific n Level

$$E = \frac{(-2.178 \times 10^{-18} \text{ J}) (Z^2)}{n^2}$$

where n is the shell # and it can be 1, 2, 3...etc

E represents the **energy** of a specific n level

Z =**atomic # (# of protons)** in an atom (ex. Hydrogen has 1 proton, $Z=1$, He^+ has 2 protons, $Z=2$)

Calculating the Energy Difference Between n Levels

$$\Delta E = -2.178 \times 10^{-18} \text{ J} (Z^2) \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where ΔE represents the **difference in energy** between 2 energy levels

Z =**atomic # (# of protons)** in an atom

n_f is referring to the **final energy level**

n_i is referring to the **initial energy level**

- If ΔE is **positive**, energy was (absorbed/released) _____ and the electron should go to a (higher/lower) _____ energy level

-
- If ΔE is negative, energy was (absorbed/released) _____ and the electron should go to a (higher/lower) _____ energy level

 WATCH OUT!

Be very careful when entering n_f and n_i values into this equation!

Example: If we are told energy is **absorbed** and an electron goes from **$n=1$ to $n=2$**

- **$n=1$ is the initial energy level, $n=2$ is the final energy level.**

Example: If we are told energy is **released** as an electron goes from **$n=3$ to $n=2$**

- **$n=3$ is the initial energy level, $n=2$ is the final energy level.**

Calculating the Wavelength of the Photon Absorbed or Emitted

Method 1:

- Use the above equation to solve for ΔE .
- Then use $E = \frac{hc}{\lambda}$ to solve for the wavelength of the photon
 - Even if ΔE is negative, use the positive value to solve for the wavelength so you **get a positive answer for wavelength** (negative wavelength doesn't make sense)

Method 2:

- Use the following equation and then take the inverse to solve for λ

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

λ =wavelength of the photon (m) -always enter answer as a positive value!

R_H =Rydberg constant= $1.097 \times 10^7 \text{ m}^{-1}$

n_i =initial energy level, n_f =final energy level

Example: Releasing or Absorbing Energy and How That Differs For Different Atoms

An electron in a one-electron species transitions from the $n = 4$ state to the $n = 2$ state.

a) Is energy released or absorbed?

b) Would more energy be absorbed/released from a sample of Hydrogen atoms or Li^{2+} ions?

Example: Excitation and Relaxation in Atoms

Calculate the wavelength, in nm, of a photon released when an electron in a hydrogen atom relaxes from $n = 3$ to $n = 1$. What is the energy of the photon if the electron is re-absorbed from $n = 1$ to $n = 3$?

Practice: Using Bohr Model Equations to Determine an Energy Level

A photon of light was released by a hydrogen atom with an energy of 1.94×10^{-18} J. If this transition ended at $n=1$, from which level did this transition begin?

☐ n=1

☐ n=2

☐ n=3

☐ n=5

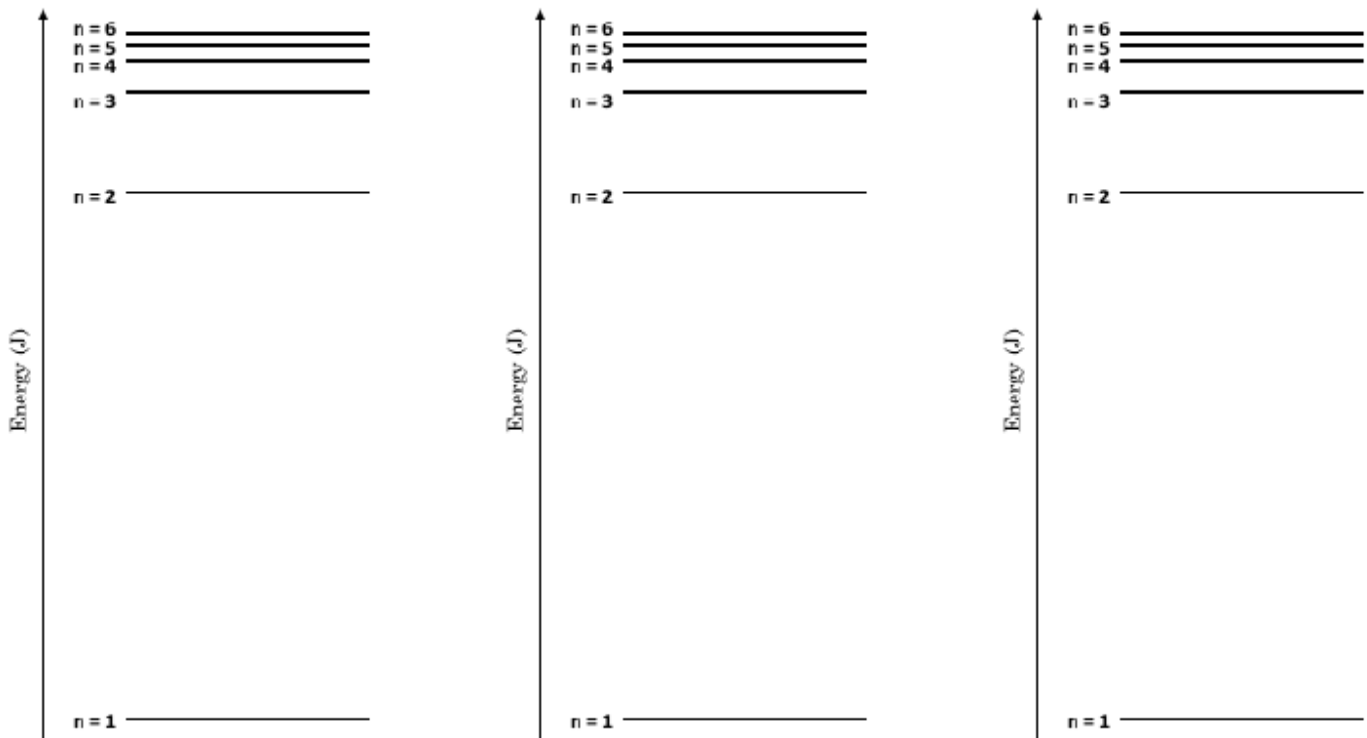
☐ n=6

2.6 How Atomic Spectra Relates to Electronic Transitions + Other Applications of the Bohr Model

2.6.1

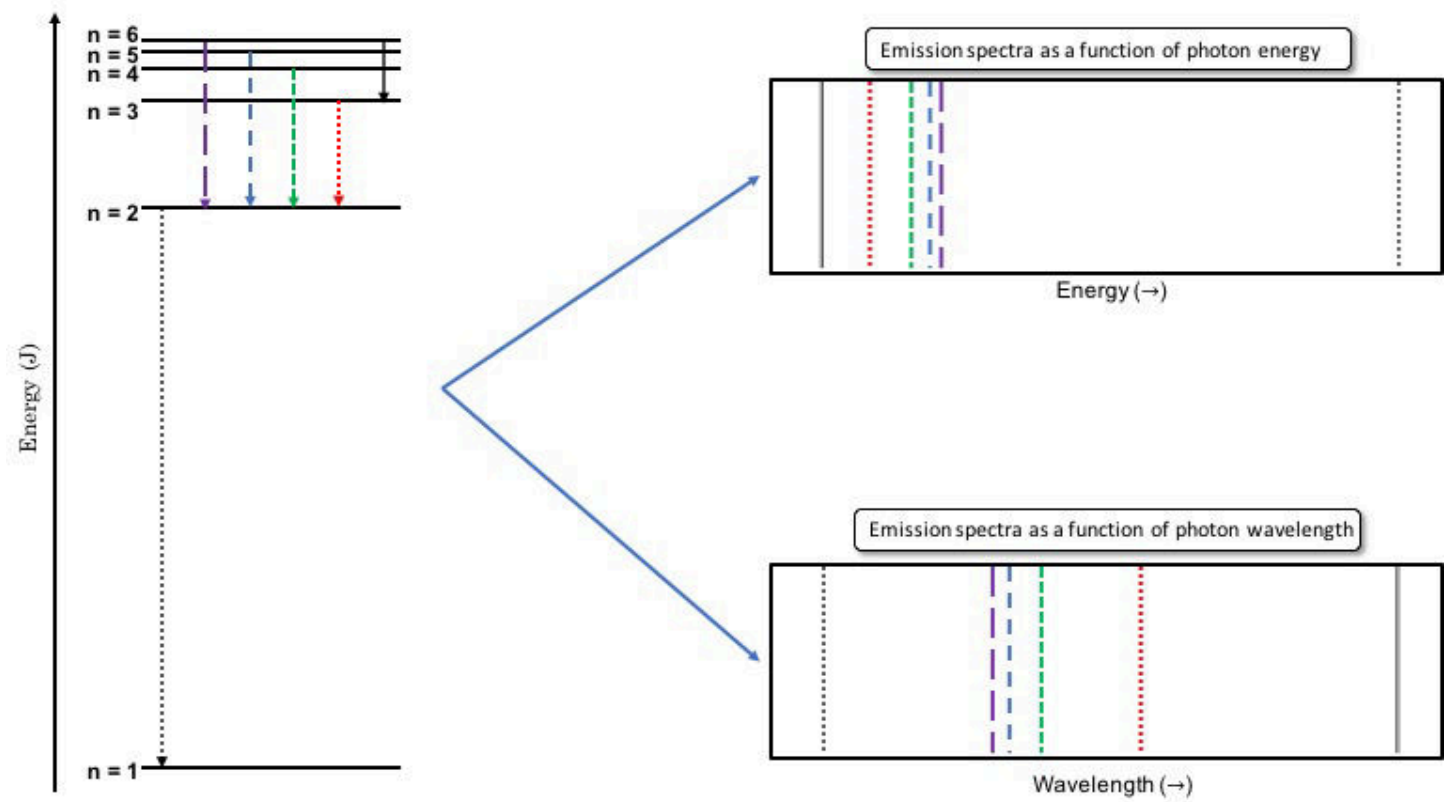
Atomic Line Spectra

Emission and Absorption



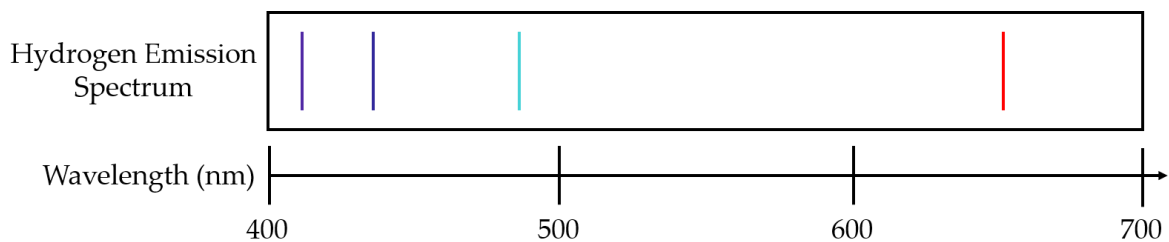
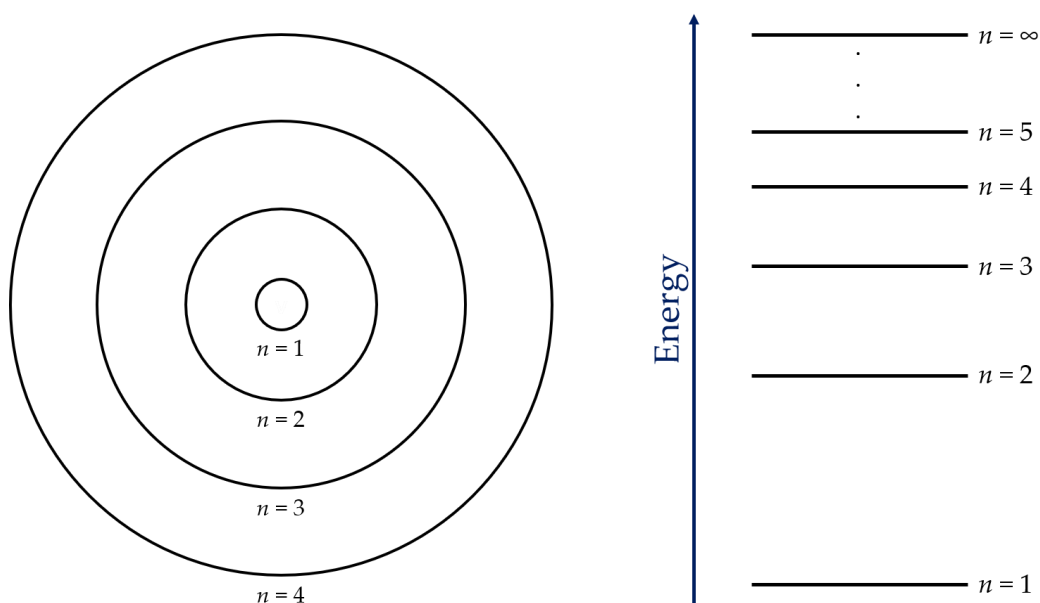
Note: $n=1$ is ground state, all other n levels are excited states

Emission Spectra



Relating Electronic Transitions to Emission Spectra

Bohr Model of the Hydrogen Atom



Hydrogen Emission Spectral Series:

Hydrogen Eimission Spectrum Series Name	n_f
Lyman	1
Balmer	2
Paschen	3
Brackett	4
Pfund	5

Example: Removing an Electron

What wavelength of light corresponds to the 4th ionization energy of Beryllium?

Can start by solving for the energy difference for this transition:

Now you can solve for the wavelength that the energy difference corresponds to:

Practice: Understanding Emissions

Which of the following transitions would emit the shortest wavelength photon?

$n=2$ to $n=1$

☐

$n=1$ to $n=6$

☐

$n=5$ to $n=1$

☐

$n=7$ to $n=2$

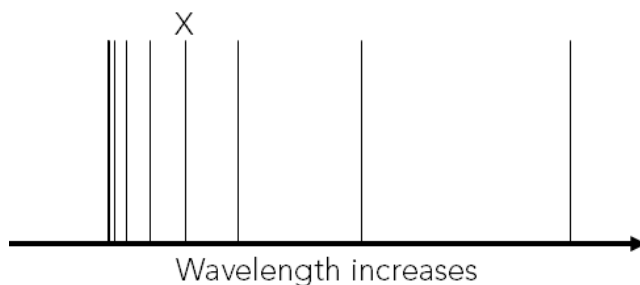
☐

$n=1$ to $n=4$

☐

2.6.5

You shine light through a sample of gaseous atomized hydrogen and take an absorption spectrum as shown below. Only transitions to or from the ground state ($n=1$) appear in the spectrum. Which of these energy-level transitions correspond to the line labelled X in the absorption spectrum?



☐ $n = 1$ to $n = 2$

☐ $n = 2$ to $n = 1$

☐ $n = 1$ to $n = 4$

☐ $n = 1$ to $n = 5$

☐ $n = 4$ to $n = 1$

☐ $n = 5$ to $n = 1$

2.7 Wave-Particle Duality & the de Broglie Equation

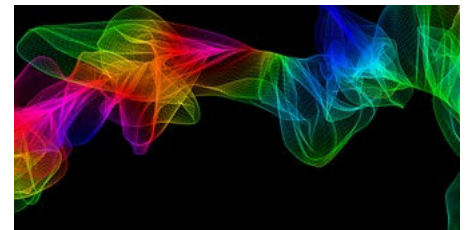
2.7.1

Wave-Particle Duality

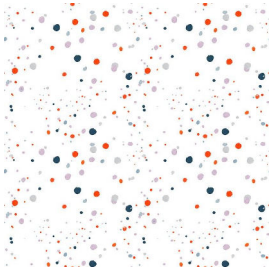
Light was originally thought of as only having wave-like properties, but was later shown to have particle-like properties as well.

Light as Waves

- When light was shown at a slit, interference patterns were recorded
- **Interference patterns** are a property of waves



Light as Particles



- Einstein showed that light can also be thought of as particles since they are like packages of exact amounts of energy
- The experiment was the **photoelectric effect** and it showed that there is **wave-particle duality** (light has both properties of waves and particles!).

What About Electrons?

Louis de Broglie thought that if light has both properties of waves and particles, electrons might also have properties of particles and waves.

He determined the following equation to explain the wave-nature of particles:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

De Broglie Equation

λ is the **de Broglie wavelength** of the small particle (**m**)

h is Planck's constant = $6.626 \times 10^{-34} \text{ Js}$ (or $6.626 \times 10^{-34} \text{ kg m}^2/\text{s}$)

m is the mass of the particle (**kg**)

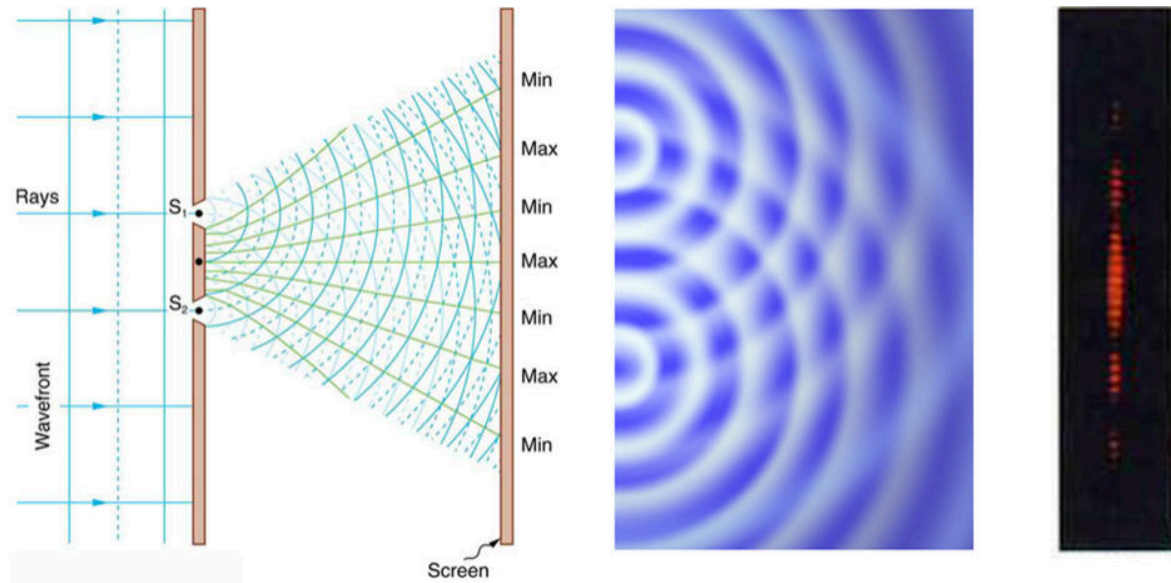
v is the velocity of the particle (**m/s**)

p is the momentum of the particle ($p=mv$)

WATCH OUT!

The v in this equation is not frequency (ν , Hz), it is velocity (v , m/s).

Light acting as a wave:



Young's double slit light experiment

Photo by Rice University / CC BY

Could electrons also act as waves?

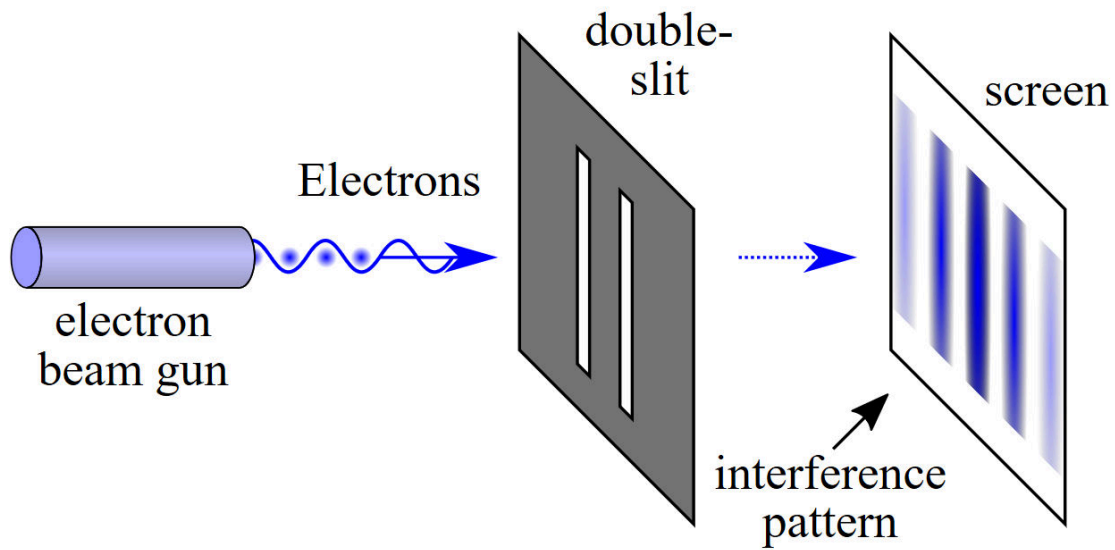


Photo by Johannes Kalliauer / CC BY

In summary

All waves can behave like particles and all particles can behave like waves! **The wave-particle duality applies to light and particles!**

Example: Wavelength of an Accelerated Neutron

At CERN, a neutron is accelerated to 2.4×10^8 m/s in a particle accelerator. The mass of the neutron is 1.675×10^{-27} kg, what is its wavelength? (h is 6.626×10^{-34} kg m²/s)

- a) 1.14×10^{-15} m
- b) 2.77×10^{-14} m
- c) 1.65×10^{-15} m
- d) 3.53×10^{-14} m

Practice: Two-Slit Experiment

If you shoot a beam of electrons at a piece of metal with two small slits in it, a diffraction pattern appears on the detector on the otherside. This is often referred to as a two slit experiment. This result shows that,

electrons can behave as particles

☐

electrons are very small

☐

electrons can behave as waves

☐

electrons can behave as both particles and waves

☐

electrons travel very fast

☐

Practice: Wave-Particle Duality

Nolan Ryan, a fastball pitcher from the 1970's, was recorded throwing a four seam fastball around 108mph (174km/h). A MLB baseball weighs approximately 145g. What is the wavelength of the baseball?



2.8 Schrodinger's Equation

2.8.1

Schrödinger Equation

$$\hat{\mathbf{H}}\psi = \mathbf{E}\psi$$

$\hat{\mathbf{H}}$: Hamiltonian

Ψ : Wavefunction

$\mathbf{E}$: Total Energy

- For all but the simplest systems this equation cannot be solved analytically (exactly) but very good approximations can be obtained by computational methods.
- The wavefunction of a system, Ψ , can be used with a variety of operators to predict properties
- While the wavefunction does not have much of a “physical meaning”, the square of the wavefunction Ψ^2 gives the probability distribution of electron density.

Ψ : Wavefunction
 Ψ^2 : Probability Distribution

2.9 Heisenberg Uncertainty Principle

2.9.1

The Heisenberg Uncertainty Principle

As electrons were discovered to have both wave and particle properties, Heisenberg found that we **cannot determine the exact position and momentum (direction and speed) simultaneously**. This is known as the Heisenberg uncertainty principle.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Δx is the **uncertainty in position**

Δp is the **uncertainty in momentum**

$\Delta p = m\Delta v$ where m is the mass (**kg**) and Δv is the uncertainty in velocity (**m/s**)

h is **Planck's constant** (6.626×10^{-34} Js or 6.626×10^{-34} kg m²/s)

The larger the value for Δx is, the (larger/smaller) _____ the value for Δp is

- In other words, a **larger value for Δx** means we are **more uncertain about the position** of the particle and **less uncertain about the momentum** of the particle!

Max Born added to this and said that we can interpret this equation to mean that we cannot say exactly where electrons are but we can say where they are likely to be.

Example: Wave Particle Duality and the Heisenberg Uncertainty Principle

A beam of electrons is accelerated towards a thin metal foil and diffracts to give the exact same pattern as a blue laser with a wavelength of 480nm. The mass of an electron is 9.11×10^{-31} kg.

a) What speed were the electrons accelerated to?

b) If we measure the speed to a precision of $\pm 1\%$ what is the minimum uncertainty in the position of the electron?

Practice: Uncertainty of Position

A circus performer is shot out of a cannon and leaves the cannon with a velocity of 6.2 ± 0.1 m/s. If the performer weighs 82 kg, what is the minimum uncertainty in their position?



6.2 m

☐

6.43×10^{-36} m

☐

8.2 m

☐

2.46×10^{-34} m

☐